MISSILE TRACKING & SECURITY SYSTEM

Ultrasonic radar turret simulation with keypad and IR access control



The working prototype with its Arduino code open on the laptop.

Project Documentation | Prepared by Kazi Jawad

1. Project Overview

The **Missile Tracking & Security System** is an Arduino-based prototype that combines two ideas in one scale model: a defense turret that scans for objects and simulates a target lock and missile launch, and an access-control system that protects a gate with a keypad password and infrared (IR) intrusion sensors. It integrates ultrasonic scanning, servo-based missile launcher simulation, IR intrusion detection, keypad password security and alarm triggers, and is designed to mimic a defense turret with access control and object detection.

The project is built as two separate Arduino sketches that share the same physical model. Each sketch can be uploaded and demonstrated on its own, which keeps the logic simple and makes every behavior easy to show and explain.

Key highlights

- **Radar scanning:** an HC-SR04 ultrasonic sensor mounted on a two-servo (X/Y) pan-tilt head, with a detection range of 2-400 cm.
- **Three-stage response:** idle, warning and target lock, driven by the measured distance.
- **Missile simulation:** two additional servos aim and "launch", followed by siren tones and an explosion effect.
- **Password gate:** a 4x4 matrix keypad with a configurable password (default `1234`).
- **Intrusion detection:** two IR sensors watch for ground objects and are disabled for 10 seconds when the correct password is entered.
- **Feedback:** LEDs, a buzzer and Serial Monitor messages (9600 baud) report every state.



Figure 1 - The physical model: radar tower, house with roof keypad, signal lights and gate arm on a landscaped base.

2. System Architecture

The system is organised into two independent subsystems. Part 1 handles detection, tracking and the simulated engagement. Part 2 handles authentication and perimeter monitoring. Both read inputs, apply simple rule-based logic and drive outputs (servos, LEDs and a buzzer).

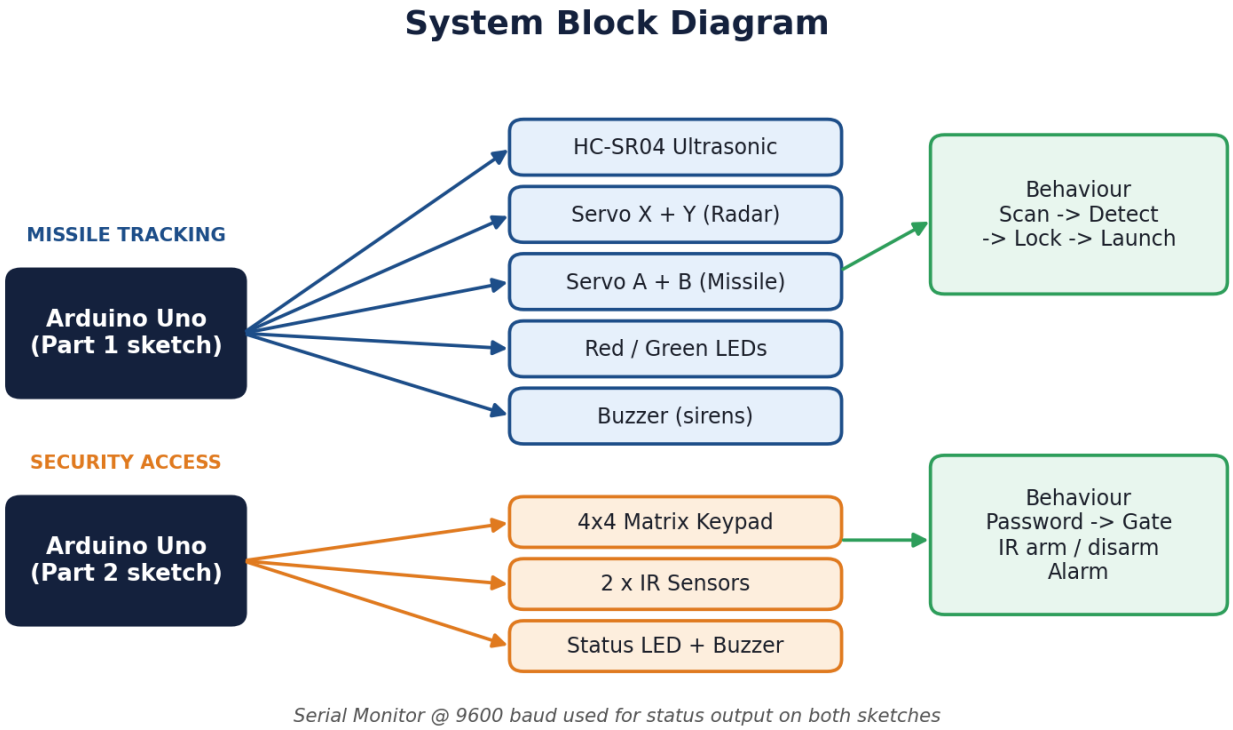


Figure 2 - Block diagram showing the two subsystems, their sensors/actuators and resulting behavior.

Project structure

Part	Name	Role
Part 1	Missile Tracking & Locking System	Ultrasonic object detection, servo radar scanning, target lock, missile launch and explosion simulation, LED/buzzer indicators.
Part 2	Security Access with Keypad & IR Sensors	Keypad password authentication, IR intrusion detection, alarm on wrong attempts, 10-second IR disable on success (gate simulation).

3. Hardware Requirements

Group	Component	Qty	Purpose
Common	Arduino Uno / Mega / Nano	1	Main microcontroller
Common	Breadboard and jumper wires	-	Prototype wiring
Part 1	Servo motors - radar (servoX, servoY)	2	Pan and tilt the ultrasonic sensor
Part 1	Servo motors - missile base (servoA, servoB)	2	Aim and launch the missile simulation
Part 1	Ultrasonic sensor (HC-SR04)	1	Distance measurement
Part 1	Red LED	1	Target lock / warning
Part 1	Green LED	1	System active / idle
Part 1	Buzzer	1	Beeps and sirens
Part 2	4x4 matrix keypad	1	Password entry
Part 2	IR sensors	2	Ground intrusion detection
Part 2	LED	1	Alert / gate status
Part 2	Buzzer	1	Alarm and gate sounds

4. Pin Configuration

Pin assignments for each sketch are listed below. Because Part 1 and Part 2 are separate sketches, the same Arduino pin numbers are reused for different components; wire the circuit that matches the sketch you are uploading.

Part 1 - Missile Tracking Pin Map

Component	Arduino Pin
Ultrasonic TRIG	**D7**
Ultrasonic ECHO	**D6**
Servo X (radar)	**D9**
Servo Y (radar)	**D10**
Servo A (missile base)	**D11**
Servo B (missile vertical)	**D3**
Buzzer	**D4**
Red LED	**D5**
Green LED	**D8**

Figure 3 - Part 1 pin map (missile tracking).

Part 2 - Keypad & IR Security Pin Map

Component	Arduino Pin
Keypad rows (R1-R4)	**D2, D3, D4, D5**
Keypad columns (C1-C4)	**D6, D7, D8, D9**
IR Sensor 1	**D10**
IR Sensor 2	**D11**
Status LED	**D12**
Buzzer	**D13**

Figure 4 - Part 2 pin map (keypad and IR security).

5. Part 1 - Missile Tracking & Locking System

In this mode the Arduino continuously sweeps the radar head using servo X and servo Y while the HC-SR04 measures the distance to whatever is in front of it. The measured distance determines how the system behaves. The Serial Monitor prints the current servo angles together with the distance, for example `X: 30, Y: 60 => Distance: 18 cm`.

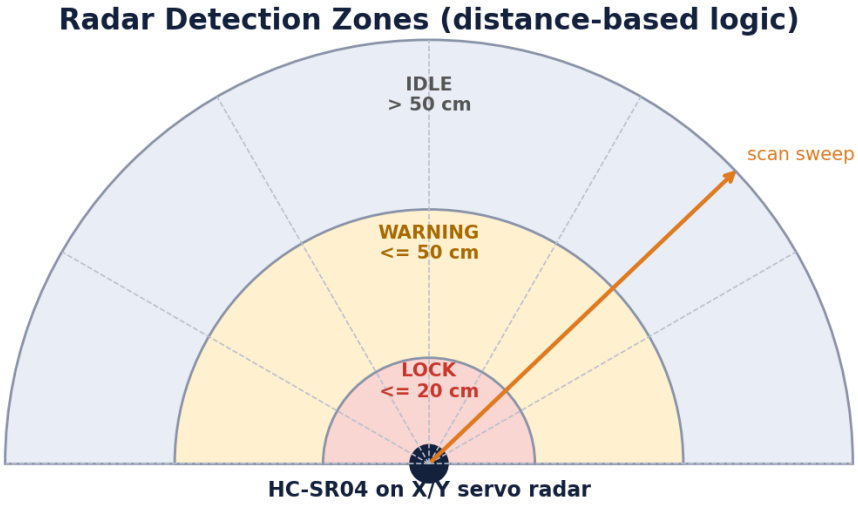


Figure 5 - Distance-based zones used by the tracking logic.

Distance	State	System response
More than 50 cm	Idle	Radar keeps scanning; green LED stays ON.
50 cm or less	Warning mode	Red and green LEDs blink alternately with warning beeps.
20 cm or less	Target lock	Red LED = lock; missile servos aim, missile launch is simulated, siren tones play, then an explosion effect.

Indicator reference

Indicator	Meaning
Green LED	System active / idle
Red LED	Target locked
Red-Green blinking	Object detected (warning)
Siren tones	Missile launch and explosion

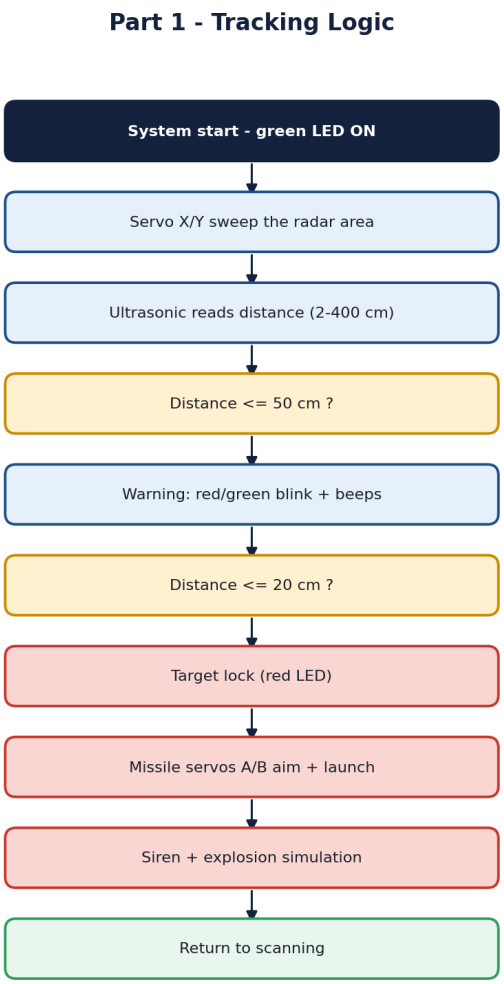


Figure 6 - Flowchart of the Part 1 tracking logic.



Figure 7 - The ultrasonic radar head on its servo pan-tilt mount, atop the tower.

6. Part 2 - Keypad & IR Security System

Part 2 simulates a protected gate. A 4x4 keypad is used to enter a password; two IR sensors watch the ground for intruders whenever they are armed. Entering the correct password temporarily disables the IR sensors so that the gate can open, while repeated wrong entries trigger an alarm.

Password system

- Default password: 1234 (configurable in the code).
- # acts as the Enter key.
- * clears the current input.

Event	System response
Correct password	Prints "Access Granted"; IR sensors are disabled for 10 seconds (gate opening), then re-activated (gate closing).
Wrong password	Prints "Wrong Password!" as a warning.
Two wrong attempts	Alarm triggers: long sirens and LED blinking.
IR detection (armed)	Ground intrusion is detected and reported while the IR sensors are enabled.

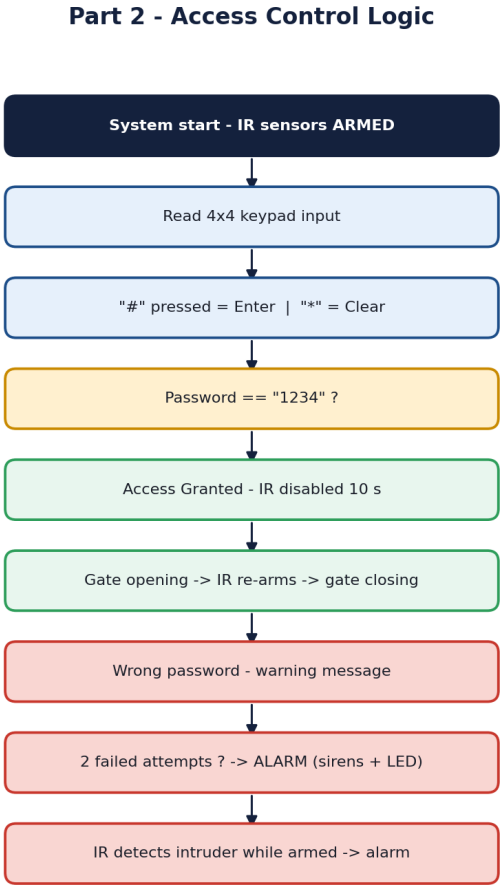


Figure 8 - Flowchart of the Part 2 access-control logic.



Figure 9 - The 4x4 keypad on the model house roof, with red/green signal lights and the servo-driven gate arm.

7. Example Serial Monitor Output

Both sketches report their state at 9600 baud. Typical output looks like this:

```
Part 1 - Tracking
Missile Tracking & Locking System Activated
X: 30, Y: 60 => Distance: 18 cm
Target lock!
Missile Launch!
Target Destroyed!
```

```
Part 2 - Access (correct password)
Access Granted
IR Sensor Disabled for 10 Seconds .. Gate Opening
IR Sensor Activating ...
Gate is closing
```

```
Part 2 - Access (wrong password)
Wrong Password!
Wrong Password!
!! ALARM TRIGGERED !!
```

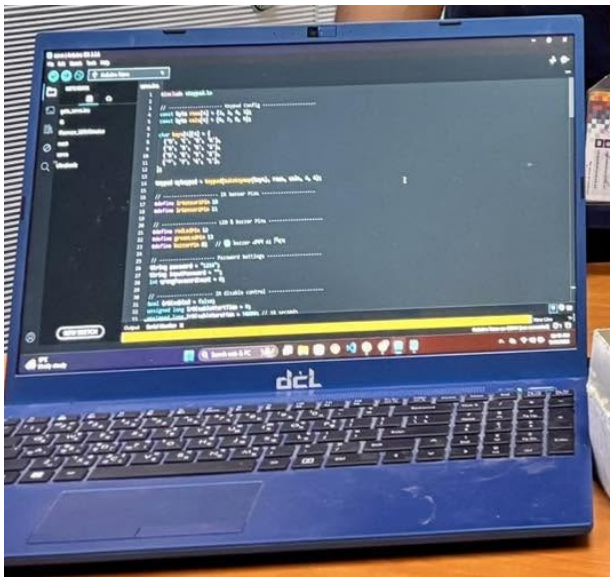


Figure 10 - The Arduino IDE with the keypad and IR sketch open on the demo laptop.

8. How to Use

- **Wire the circuit** as shown in the pin maps for the sketch you want to run.
- **Part 1:** upload the Missile Tracking code to the Arduino for the tracking demo.
- **Part 2:** upload the Keypad & IR code for access control and security.
- **Monitor:** open the Serial Monitor at 9600 baud to follow the output.
- **Test Part 1:** place an object in front of the radar at more than 50 cm, then at 50 cm or less, then at 20 cm or less to see idle, warning and lock states.
- **Test Part 2:** enter `1234#` to open the gate, then enter a wrong code twice to trigger the alarm.

9. Future Improvements

- **Bluetooth / WiFi control** for remote access and monitoring.
- **LCD display** for on-device status updates.
- **Motion-tracking camera** (OpenCV + Arduino) for advanced targeting.
- **EEPROM support** to save the password permanently.

10. Conclusion

The Missile Tracking & Security System demonstrates how a few low-cost components - an ultrasonic sensor, servos, IR sensors, a keypad, LEDs and a buzzer - can be combined to model both detection/response and access control. Its distance-based tracking logic, password-protected gate and clear audio-visual feedback make it a practical, easy-to-demonstrate platform for learning embedded systems, sensor interfacing and state-based control, with plenty of room to grow through the improvements listed above.

Enjoy building your Arduino Missile Defense + Security System!